VEXON DEVELOPER HANDBOOK · GUIDE 06

Debugging and Error Recovery Guide

Find where a VEXON problem starts, prove the cause, fix it safely, and stop it coming back, without depending on AI

Written for	A student maintaining VEXON alone who needs a repeatable way to investigate errors
Project	VEXON / CampusConnect (repository: this folder's parent project)
Evidence	Generated from a direct inspection of the repository source code
Repository state	branch fix/csrf-legacy-cookie, commit af623fb, plus 126 uncommitted working-tree change(s)
Generated	16 October 2026, 21:11
Rebuild	python docs/developer-handbook/build_handbook.py

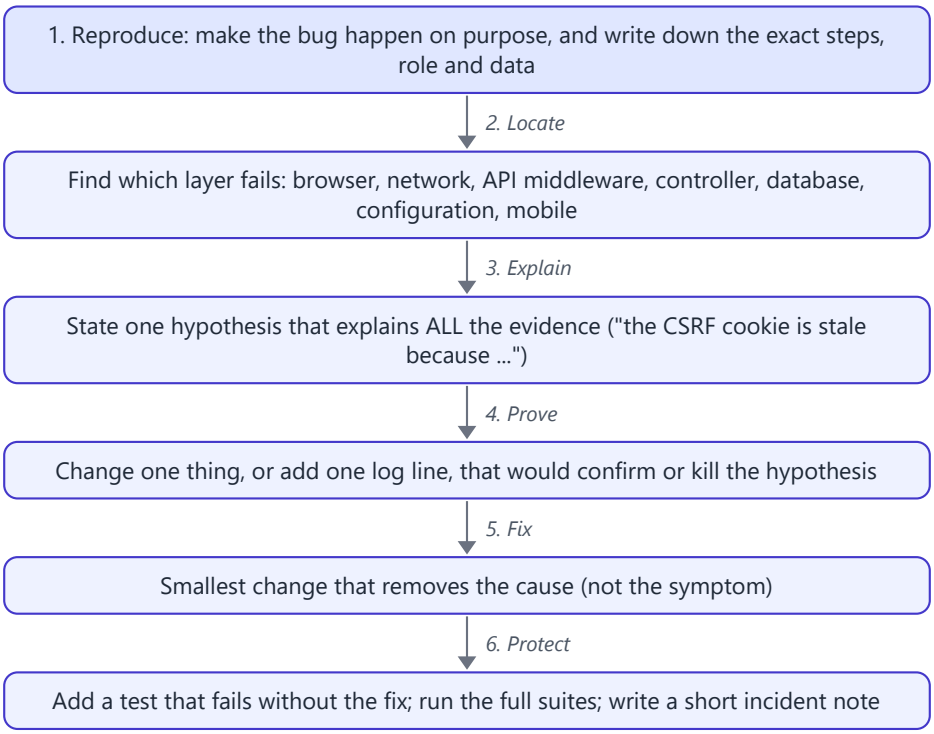
Accuracy rule: every statement about VEXON in this guide is based on files in the repository. Anything that could not be confirmed is labelled 'unverified', 'planned' or 'hypothetical'. No secrets, passwords or real student data are reproduced.

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The debugging mindset

Debugging is **collecting evidence until only one explanation is left**. It is not guessing. Every investigation in VEXON follows the same loop:



> **Key idea**
> The most valuable sentence in any bug report is the **exact error message** together with **where you saw it** (browser console, Network tab response, server terminal, Android logcat). Copy it exactly. Do not paraphrase.

Where evidence lives

Source	How to open it	What it tells you
Browser Console	F12, then Console	JavaScript exceptions with file and line; React warnings
Browser Network tab	F12, then Network; filter `Fetch/XHR`	Every API call: method, URL, status, request headers, response body
Browser Application tab	F12, then Application, then Cookies	`cc_refresh` (httpOnly) and `cc_csrf` cookies in development; `__Secure-cc_refresh` and `__Host-cc_csrf` in production
React DevTools / Redux DevTools	Browser extensions	Component props and state; the RTK Query cache and `auth` state
Server terminal (`npm run dev`)	The window running the API	`morgan` request lines; `[error]` stack traces; `[db]`, `[env]`, `[socket]` messages
MongoDB	mongosh or Compass, on **dev** data	Whether the document really exists or changed
Admin, then Security & Logs	Web, admin only	Sign-in failures, session revocations, scope refusals, OTP events
Android logcat	Android Studio Logcat, or `adb logcat`	Kotlin exceptions, OkHttp failures, crashes
GitHub Actions	The repository's Actions tab	CI test and build failures

Source	How to open it	What it tells you
Render dashboard	Service, then Logs	Production server logs

Browser developer tools in practice

Console errors

A red error such as `TypeError: Cannot read properties of undefined (reading 'map') at Notifications (Notifications.jsx:50)` tells you three things: **what** happened (reading `.map` of `undefined`), **where** (the file and line), and **what to suspect** (data not loaded yet, or the API returned another shape). In VEXON the usual fix is optional chaining (`data?.items?.length`) plus a loading state. Check `Notifications.jsx` to see how the real code avoids this.

The Network tab

For a failing action:

1. Click the red request.
2. **Headers**: check the request URL and method, and that `Authorization: Bearer ...` and `x-csrf-token` are present.
3. **Response**: read the JSON `message`. VEXON always sends one.
4. **Status**: use the table below.
5. **Timing**: a very long wait followed by failure suggests the server is asleep (Render free plan, about 30 s to wake) or there is a network problem.

Status	First suspicion in VEXON	Where to look
(failed) / `FETCH_ERROR`	Server not running, wrong port, CORS, or offline	Server terminal; `client/vite.config.js` proxy
401	Access token expired and the refresh failed; session revoked; idle timeout (`reason: 'idle'`)	`server/src/middleware/auth.js`, `server/src/utils/sessionIdle.js`, `client/src/services/api.js`
403 "Invalid or expired CSRF token"	Missing or stale CSRF cookie or header	`server/src/middleware/security.js`, `client/src/services/csrfCookie.js`
403 "Cross-origin request blocked by security policy"	The page's origin is not in `CLIENT_URL` / `APP_URL`	`server/.env` `CLIENT_URL`; `security.js` `getAllowedOrigins`
403 "You do not have permission ..."	`authorize(...)` refused the role	The route in `server/src/routes/index.js`
403 "The ... can only use ..."	`confinedRoleGuard` (coe, office, transport)	`server/src/middleware/coeGuard.js`
404 "Route not found: ..."	Wrong URL or method, or a route that is not registered	`routes/index.js`; RTK endpoint URL in `api.js`
409	Duplicate unique value, or a business conflict (e.g. an exam blocks attendance)	`middleware/error.js`; the controller
422	Validation: read `errors[].field`	The `body(...)` rules for the route; Mongoose schema
423	Account locked after failed sign-ins	`authController.login`; wait, or have an admin reset the password
429	Rate limiter hit; the message names which one	`server/src/middleware/rateLimit.js`
500	A bug or an unexpected exception	Server terminal `[error]` stack trace

Reading a stack trace

> **Evidence status**
> The trace below is a **hypothetical example**, written to teach the method. It is not a recorded VEXON error.

```
[error] TypeError: Cannot read properties of null (reading 'department')
    at createComplaint (file:///.../server/src/controllers/complaintController.js:126:34)
    at file:///.../server/src/utils/http.js:13:19
    at Layer.handle [as handle_request] (.../node_modules/express/lib/router/layer.js:95:5)
    at next (.../node_modules/express/lib/router/route.js:149:13)
```

- **Line 1**: the error type and message. Here, something expected to be an object was `null`.
- **Read downward and stop at the first line inside VEXON's own code**, not `node_modules`. Here it is `complaintController.js:126`, column 34.
- `http.js:13` is `asyncHandler`. It appears in almost every trace and is rarely the cause.
- Open line 126, find which variable could be `null`, and ask *why*. A missing populate? A user without a department? Then reproduce it with that data.

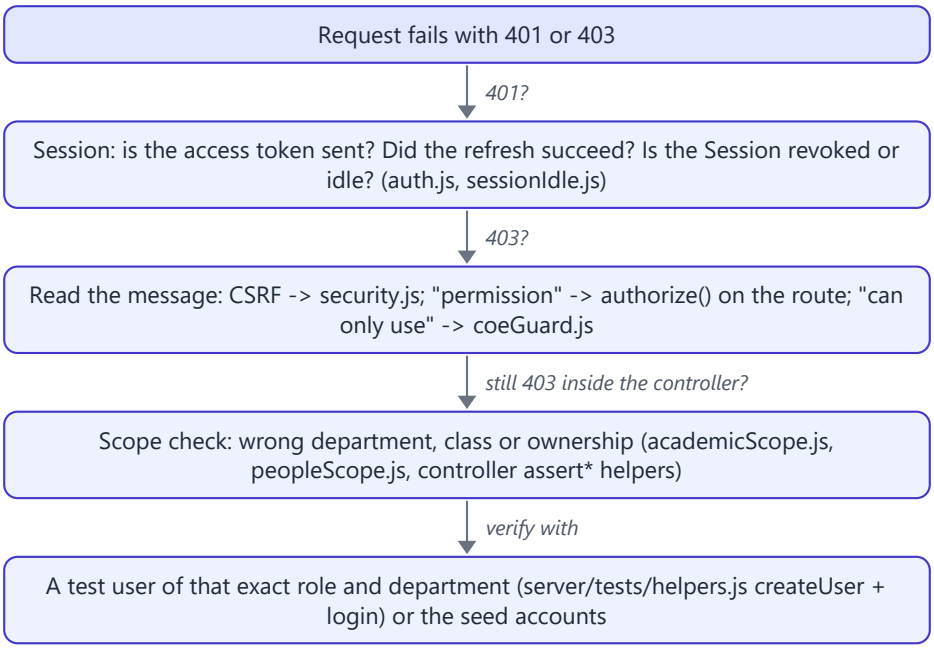
Common errors and their usual causes

Error text (where)	Usual cause	Fix direction
`Cannot read properties of undefined` (console or terminal)	Data not loaded, a missing field, or a wrong shape	Optional chaining plus a loading or empty state; fix the data source
`X is not a function`	Wrong import (default vs named), or a typo	Check the `export` in the source file
`Objects are not valid as a React child`	Rendering an object in JSX	Render a field: `{user.name}`
`Each child in a list should have a unique "key"`	`.map()` without `key`	`key={item._id}`
`Too many re-renders`	Calling a state setter during render	Move it into an event handler or effect
`Cannot find module '../x'` / `ERR_MODULE_NOT_FOUND` (server)	Missing `.js` extension in a server import, or a wrong path	Server ES-module imports need `.js`
`EADDRINUSE :::5000`	Another process is using port 5000	Stop the other process, or set `PORT` (README section 7)
`[db] Could not connect to MongoDB (local)`	MongoDB service stopped; wrong `MONGO_URI`	Start the service (`net start MongoDB`, as admin, on Windows)
`DB_TARGET=atlas but MONGO_URI_ATLAS is not set in . env`	Switched to Atlas without a URI	Set it, or use `DB_TARGET=local`
`Missing required environment variable: JWT_ACCESS_SECRET` (production)	Secret not configured on the host	Set it in the host's environment (Render dashboard)
`[env] ... using an insecure development default`	No secret in a dev `.env`	Fine for local only; never in production
`Refusing to run tests against a non-test database`	`TEST_MONGO_URI` does not end in `_test`	Use `.../cmp_test`
`MongoServerError: E11000 duplicate key` (409 to the client)	A unique index was violated	Show a friendly message; check whether duplicates are legitimate

Database connection and validation errors

1. Is MongoDB running? On Windows, open `services.msc` and find "MongoDB Server"; or run `mongosh` and see whether it connects.
2. Which database is the server using? The start-up line `[db] MongoDB connected (local) -> host:port/name` tells you. A surprising name means a surprising `.env`.
3. Validation errors (422) list `field` and `message`. Compare them with the schema in `server/src/models/<Model>.js` and the route's `body(...)` rules.
4. Unique index failures at start-up (from `syncIndexes`) mean existing data breaks a new rule. Find the duplicates with an aggregation (Guide 04).

Authentication and permission failures



Common mistake

"Fixing" a 403 by adding a role to `authorize(...)` without asking whether that role *should* have access. A 403 is often the system working correctly. Check `server/src/constants.js` and the module's rules in Guide 05 first.

Environment and configuration problems

Symptom	Configuration to check
Web works, but writes fail with a 403 origin error after deploying	`CLIENT_URL` and `APP_URL` must equal the real public URL (`render.yaml` explains why)
Socket "Reconnecting..." forever in production	Same `CLIENT_URL` (Socket.IO CORS), plus a proxy that supports WebSockets
Every user locked out at once in production	`TRUST_PROXY` wrong, so all visitors share the proxy IP (`trustProxySetting` in `server/src/app.js`)
Uploaded images vanish after a Render deploy	Cloudinary not configured; files went to local disk or the database
Bus map is blank	`MAP_STYLE_URL` unreachable, or its origin missing from the CSP (`MAP_CSP_ORIGINS`)

Symptom	Configuration to check
Parent OTP never arrives	`OTP_SMS_ENABLED` and the `SMS_*` variables; with SMS off, admins read OTPs in Security & Logs, then OTP
Study Assistant says it is unavailable	`GROQ_API_KEY` is unset
Android app cannot reach the server	The server URL set on the login screen; `10.0.2.2` only works in the emulator; plain HTTP needs an `http://` build setting

Never print secret values while debugging. Print `Boolean(process.env.GROQ_API_KEY)`, not the key itself.

Socket.IO troubleshooting

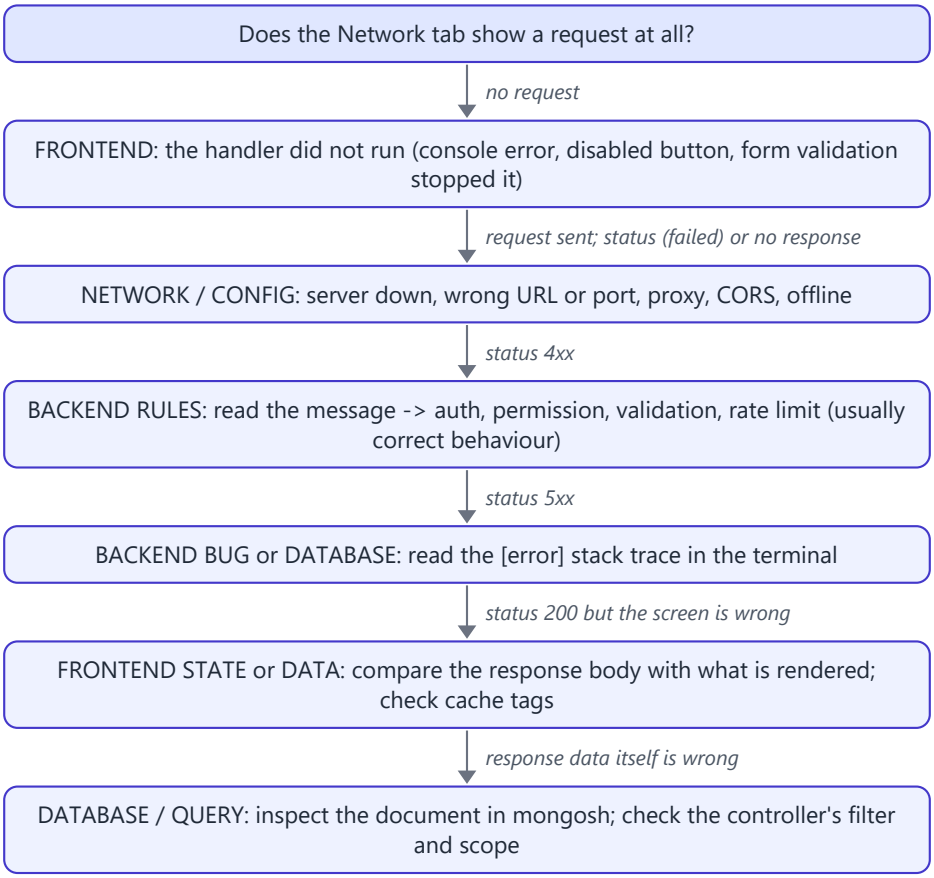
1. Network tab, filter **WS**: is there a `socket.io` connection with status 101?
2. Console: `connect_error` with `unauthorized` means the access token was rejected at the handshake. `client/src/services/socket.js` then refreshes and retries.
3. Rooms are joined **after a database check**. If a chat updates for one user but not another, check membership (`canUseConversation` in `server/src/config/socket.js`).
4. After a reconnect, rooms are re-joined by `connectHooks` (web) or the Kotlin `Realtime` equivalent. A missing re-join means stale data until reload.
5. A server-side live check: `node server/scripts/realtime-smoke.mjs` (needs a running server and the seed accounts).

Android build and runtime errors

Problem	Evidence	Usual fix
Gradle sync or build fails: "Unsupported class file major version" / toolchain	Build output	The project uses `jvmToolchain(17)`. Install JDK 17, or let Gradle provision it
`SDK location not found`	Build output	Create `mobile-kotlin/local.properties` with `sdk.dir=...` (never commit it)
App shows "Cannot reach the JNN server"	The app's `errMsg` (`core/Http.kt`)	Wrong server URL; server down or asleep; phone not on the same network
`CLEARTEXT communication not permitted`	Logcat	Release build pointed at `http://`; build with an `http://` `vexon.apiUrl` or use HTTPS
Crash on a screen	Logcat `FATAL EXCEPTION` with a `com.vexon.campus...` frame	Open that file and line; often a JSON field is missing (see `core/Json.kt` helpers)
Push never arrives	Server logs; `DeviceToken` collection	`google-services.json` missing at build time (push disabled); `FCM_SERVICE_ACCOUNT` unset on the server
Debug app shows production data	`gradle.properties` `vexon.apiUrl`	Build with `-Pvexon.apiUrl=http://10.0.2.2:5000` for a local server

Useful commands, run from `mobile-kotlin/`: `gradlew.bat assembleDebug`, `gradlew.bat testDebugUnitTest`, and `adb logcat | findstr vexon` (Windows).

Identifying the layer: frontend, backend, database, network or configuration



Reproducing and minimising a bug

1. Write the steps as a numbered list: role, starting page, clicks, inputs, expected result, actual result.
2. Remove steps one by one while the bug still happens. That gives a **minimal reproduction**.
3. Turn the reproduction into **a request**, a `curl` command or a `ctx.request(...)` call in a test, so it runs without clicking.
4. If it depends on data, create that data in the test (`ctx.createUser({ role: 'hod', department: 'CSE' })`). **Do not depend on real records.**

Evaluating an AI-generated fix

Before accepting any fix, from Claude or anyone else, answer these:

- [ ] Does the explanation match **all** the evidence, including the exact error message and the file and line?
- [ ] Does the change touch only the files needed? Read the whole `git diff`.
- [ ] Does it remove a **security check** (`authorize`, a scope assert, CSRF, validation, rate limit) or widen a role list? If so, why is that correct?
- [ ] Does it change an **API contract** (URL, request or response shape) used by web, Kotlin, Expo or iOS?
- [ ] Does it hide the error instead of fixing it (an empty `catch {}`, `|| []` everywhere, a deleted test)?
- [ ] Is there a new or updated **test** that fails without the fix?

- [ ] Do the full suites still pass (`npm test --prefix server`, `npm test --prefix client`, Kotlin `testDebugUnitTest`)?
- [ ] Did you run the real flow once, as the affected role?
- [ ] Does it introduce a dependency, secret, or hard-coded URL?

> **Warning**
> Never paste real student data, `.env` values, tokens or database URIs into an AI prompt. Replace them with fake values first.

Testing a fix and preventing regressions

1. Write a test that reproduces the bug, and **watch it fail**.
2. Apply the fix and watch the test pass.
3. Run the whole suite. A fix in shared code (`utils/`, `middleware/`, `constants.js`) can break other modules.
4. Run the client build if you touched `client/`.
5. Note the bug and the test name in your commit message, so the history explains the "why".

Guide 07 shows how to write tests in VEXON's style.

Troubleshooting table

> **Evidence status**
> Rows are **realistic scenarios derived from VEXON's architecture**. They are not records of incidents that occurred. File names are real.

Symptom	Evidence to collect	Likely causes	Files or components to inspect	Safe next step	How to verify
Web user logged out after about 2 minutes	Response `reason: 'idle'`; role of the user	Web idle rule for non-exempt roles (by design); activity pings not sent	`server/src/utils/sessionIdle.js`, `client/src/components/layout/useIdleSession.js`, `client/src/services/idleTracker.js`	Confirm the role is meant to be non-exempt; check that `POST /api/auth/activity` is sent on interaction	`server/tests/idle-session.test.js`; manual test with that role
Every write fails with 403 CSRF	Response message; Application tab cookies	Stale legacy cookie; a cookie blocked; the header not sent	`server/src/middleware/security.js`, `client/src/services/csrfCookie.js`, `client/src/services/api.js`	Sign out and in again (issues a fresh pair); compare cookie and header	`csrf-legacy-cookie` and `csrf-self-heal` tests
Login always "Invalid email or password" for a real user	`SecurityLog` / `LoginHistory` reason	Wrong password; account inactive (that would be a 403); email case or typo	`authController.login`, `models/User.js`	Admin resets the password (`POST /api/users/:id/reset-password`)	User signs in; `LOGIN_SUCCESS` in Security & Logs
HOD cannot edit a timetable slot	403 or 422 message text	Slot outside the HOD's department or years; first-year rule	`timetableController.js`, `constants.js` (`isFirstYearHod`)	Check the HOD's `department` and the class's year	`timetable-roles` test; retry as the correct HOD

Symptom	Evidence to collect	Likely causes	Files or components to inspect	Safe next step	How to verify
Faculty cannot mark attendance today	409 or 422 message	Holiday declared; published exam overlaps the period; not the faculty's scheduled class	`attendanceController.markAttendance`, `holidayController.js`, `examController.js`	Read the message; check Holidays and the Exam timetable	`attendance-timetable` and `exam-timetable` tests
Gate pass cannot be forwarded to the HOD	Message "Verify the parent's OTP before forwarding"	Parent OTP not verified yet	`gatePassController.js` (`facultyReview`), `services/otp/otpService.js`	Request and verify the OTP first; if SMS is off, an admin reads it in Security & Logs	`outpass-otp` test; pass reaches `pending_hod`
"Send OTP" fails with "not configured"	Response message; server log	`OTP_SMS_ENABLED=true` without the `SMS_*` variables	`server/src/config/env.js`, `services/otp/providers/collegeSmsProvider.js`	Set the variables on the host, or turn SMS off	`otp-send-error-message` test; a real send on staging
Bus not shown on the map	Bus `lastLocation`; trip status; server rejections	No accepted fix yet; a fix rejected (stale, inaccurate, jump); the viewer may not see that bus	`services/transport/ingest.js`, `access.js`, `client/src/pages/transport/Transport.jsx`	Run `simulate-bus.mjs` on **dev** with a test bus	`transport.test.js`; marker appears in dev
Notification arrives in the app but not as a phone push	`Notification.delivery` field; server log	FCM not configured; device token missing or expired	`server/src/services/pushService.js`, `fcm.js`, Kotlin `push/Push.kt`	Check `FCM_SERVICE_ACCOUNT` (presence only) and the `DeviceToken` rows	`notifications-push` test; a real device test
Chat messages appear only after a reload	WS frames; console `connect_error`	Socket not connected; room not joined; access token expired at handshake	`client/src/services/socket.js`, `server/src/config/socket.js`	Check the Live indicator and the WS tab; reconnect	`chat*` tests; `realtime-smoke.mjs`
Uploaded file rejected with 415 or 400	Response message	Real type (magic bytes) not allowed; too big	`server/src/utils/storage.js`, `MAX_UPLOAD_MB`	Try a genuine PNG or PDF under 5 MB	`input-hardening` test; upload succeeds
Server will not start in production	Render logs	A missing secret throws; MongoDB unreachable	`server/src/config/env.js`, `config/db.js`	Compare the variables with `server/.env.example` (names only)	`/api/health` returns `{status: 'ok'}`
Study Assistant errors	Response status and message	Groq key missing; rate limit; provider outage	`server/src/services/ai/groqService.js`, `rateLimit.js` (AI limiters)	Check that the key exists; wait for the limit window	`study-assistant` test; one real question
Kotlin app stuck on "Can't reach server"	Login screen server value; logcat	Wrong URL; server asleep; cleartext blocked	`core/Config.kt`, `core/Http.kt`, `app/build.gradle.kts`	Use "Test & save" on the server address	App signs in

Repeatable debugging checklist

- [ ] I can reproduce it, and I wrote the steps, role and data.
- [ ] I copied the exact error message and where it appeared.
- [ ] I know the request (method, URL, status, response `message`).
- [ ] I identified the layer (frontend, network, backend rules, backend bug, database, configuration, mobile).
- [ ] I found the exact file and line (stack trace or a message search).
- [ ] I have one hypothesis that explains all the evidence.
- [ ] I proved it with a single change or log.
- [ ] My fix is the smallest one that removes the cause.
- [ ] A test fails before the fix and passes after.
- [ ] Full server and client suites pass; the client build passes; Kotlin tests pass if I touched Android.

- [ ] I removed temporary logs and reviewed `git diff`.
- [ ] I wrote an incident note if users were affected.

Incident report template

```
INCIDENT REPORT - VEXON
Title:
Date/time detected:            Reported by:
Environment:  local / staging / production (Render)    Version (git commit):
Affected users / roles / modules:
Impact (what users could not do, for how long):

Timeline:
  - hh:mm  first report
  - hh:mm  investigation started
  - hh:mm  cause identified
  - hh:mm  fix deployed / workaround applied

Evidence (exact messages, status codes, log lines - NO secrets, NO personal data):

Root cause:
Fix (commit / files changed):
Verification (tests run + manual check, with results):
Prevention (new test, monitoring, documentation):
Follow-up actions and owners:
```

Source references

server/src/middleware/auth.js · server/src/middleware/security.js · server/src/middleware/coeGuard.js · server/src/middleware/rateLimit.js · server/src/middleware/error.js · server/src/config/env.js · server/src/config/db.js · server/src/config/socket.js · server/src/app.js · server/src/utils/sessionIdle.js · server/src/utils/storage.js · server/src/controllers/authController.js · server/src/controllers/gatePassController.js · server/src/services/transport/ingest.js · server/src/services/pushService.js · server/scripts/realtime-smoke.mjs · server/scripts/simulate-bus.mjs · client/src/services/api.js · client/src/services/socket.js · client/src/services/csrfCookie.js · client/src/components/layout/useIdleSession.js · mobile-kotlin/app/src/main/java/com/vexon/campus/core/Http.kt · mobile-kotlin/app/src/main/java/com/vexon/campus/core/Config.kt · mobile-kotlin/app/build.gradle.kts · mobile-kotlin/gradle.properties